

Tzu-Wei Wang (William), Mr.

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RESEARCH FOCUS

characterization-driven materials research (microscopy/thermal/mechanical) + data/ML for energy & extreme-environment applications (electrochemistry, HEAs/SRMs, fusion materials).

EDUCATION

Nanyang Technological University (NTU), Singapore

Aug 2023 - Present

Bachelor in Materials Science and Engineering and Data Analytics (Double Major)

- GPA: 4.74/ 5.00 (Dean's Distinction Award, AY2023/24)

RESEARCH EXPERIENCE

Research Intern, Emerging nanoscience Research Institute (EnRI), Singapore

May – July 2026

Characterization-focused internship on HEA-based SRMs: XRD/SEM/(EBSD) microstructure; DSC/TGA & bomb calorimetry (energy release); compression/hardness + nanoindentation analysis.

Research Intern, Forschungszentrum Jülich, Germany

Jun - Aug 2025

Investigated how TEM imaging parameters affect structural contrast in materials. Used Dr. Probe to simulate TEM images and distinguish true microstructural features from imaging artifacts.

Undergraduate Research Experience on Campus (URECA) project, NTU, Singapore

May 2025 - Present

Collected and processed SEM/TEM images with controlled imaging parameters under Prof. Eason (Yi-Sheng) Chen and Dr. Gerfried Millner. Applied machine learning for structure–property analysis of stainless steels.

URECA project, NTU, Singapore

Oct 2023 - Mar 2025

Conduct experiments on transition metal electrocatalysts for H₂O₂ electrosynthesis. Perform electrochemical characterization (CV, LSV, RRDE) to evaluate activity and selectivity. Performed limited ML analysis for data interpretation.

CONFERENCE PRESENTATIONS

Wang, T.-W.[†], & Millner, G.[†] (2026). *"Quantitative Microscopy and Informatics for Property Prediction in Commercial Stainless Steels"*. Abstract accepted to Microscopy & Microanalysis (M&M 2026), Milwaukee, WI, USA

Wang, T.-W.[†], & Millner, G.[†] (2026). *"Explainable inverse design of battery materials via multi-model learning and conditional filtering"*. Presented at The Minerals, Metals & Materials Society (TMS 2026) Annual Meeting, San Diego, CA, USA.

Wang, T.-W., & ZengYu, H. (2025). *"Machine learning assisted mapping research of electrocatalysts study for hydrogen peroxide electrosynthesis"*. Oral & poster presentation, International Conference of Undergraduate Research (ICUR 2025), Singapore.

Millner, G.[†], & **Wang, T.-W.[†]** (2025). *"Explainable AI for Property Prediction in Scrap-Based Steel Production: A Public Data Implementation"*. Accepted conference paper, European Steel Technology and Application Days (ESTAD 2025), Verona, Italy.

MANUSCRIPT IN PREPARATION

Zengyu, H.[†], **Wang, T.-W.[†]**, et al. (2025). *"Machine learning assisted mapping research of electrocatalysts study for hydrogen peroxide electrosynthesis"*

Wang, T.-W., Millner, G. (2026). *"Prototype-Constrained Inverse Design of Li-Ion Electrode Materials with Multi-Tier Stability Validation"*

Millner, G., **Wang, T.-W.**, et al. (2026). *"Explainable AI for process–structure–property prediction in metallic materials"*

AWARDS & FUNDING

Gold Medal, International Genetically Engineered Machine (iGEM) Competition

Jan - Dec 2023

NTU President Research Scholar, URECA Programme

Mar 2025

3rd Place, URECA Poster Competition, College of Engineering, NTU

May - Aug 2025

Students' Global Experience (SGE) Award, NTU — USD 1,560

Jun - Aug 2025

International Research Funding Program (MOE, Taiwan) — USD 7,500

Dec 2023

National Tsing Hua University Research Grant — USD 7,190

Dec 2023

PROFESSIONAL & LEADERSHIP

Trainee, Graduate Research Innovation Programme (GRIP) Phase I, NTU	Jan - May 2025
Vice President, ACS Student Chapter (NTU)	2025 - Present

SKILLS

TEM/STEM, SEM, XRD, EBSD; Python, PyTorch; CHGNet, MACE, Dr. Probe; English, Chinese.